

Supplementary Material for “OctaOctree Neural Radiosity for Real-time Glossy Material Rendering in Static Scenes”

JIERUI REN, Peking University, China
HAOJIE JIN, Peking University, China
BO PANG, Peking University, China
MENG GAI, Peking University, China
FEI ZHU, Peking University, China
YISONG CHEN, Peking University, China
SHENG LI*, Peking University, China

CCS Concepts: • **Computing methodologies** → **Ray tracing**; **Neural networks**.

Additional Key Words and Phrases: Neural scene representation, global illumination, neural radiosity, glossy materials

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A Representation Analysis

This section provides detailed visualizations and quantitative results for the representation analyses summarized in the main paper.

Feature visualization. We visualize the learned features at different OctaOctree levels for a representative scene (see Figure 1). The visualization reveals the complementary spatial-angular decomposition described in the main paper: coarse spatial levels capture broader reflected structures with stronger directional selectivity and low-frequency spatial patterns, while finer spatial levels focus on localized high-frequency detail and short-range illumination variation. These observations further suggest that the hierarchy organizes outgoing radiance according to the spatial-angular reuse pattern of glossy transport, rather than merely increasing representation capacity.

*Corresponding author.

Authors' Contact Information: Jierui Ren, 2201112464@stu.pku.edu.cn, Peking University, Beijing, China; Haojie Jin, 2201111642@pku.edu.cn, Peking University, Beijing, China; Bo Pang, bo98@stu.pku.edu.cn, Peking University, Beijing, China; Meng Gai, gaimeng@pku.edu.cn, Peking University, Beijing, China; Fei Zhu, feizhu@pku.edu.cn, Peking University, Beijing, China; Yisong Chen, chenysisong@pku.edu.cn, Peking University, Beijing, China; Sheng Li, lisheng@pku.edu.cn, Peking University, Beijing, China.



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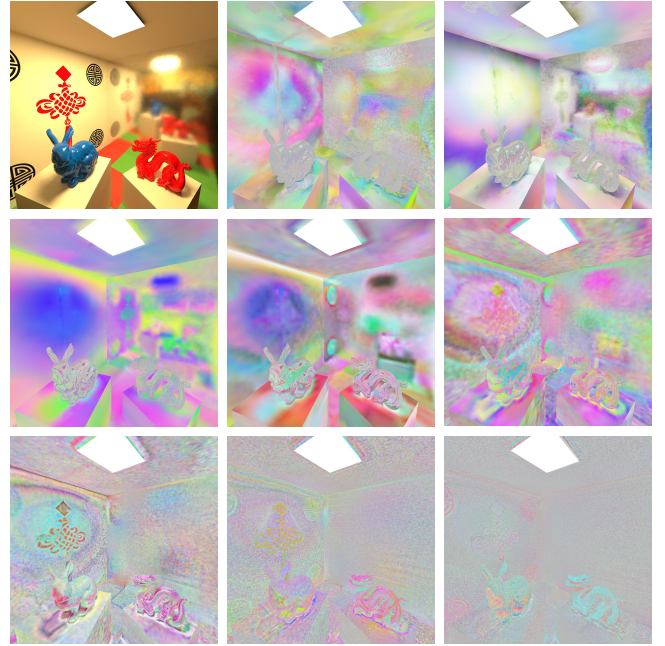


Fig. 1. Visualization of learned features at different OctaOctree levels. Coarse spatial levels capture broader reflected structures with stronger directional selectivity, while finer spatial levels focus on local high-frequency changes, geometric detail, and short-range illumination variation.

Structure statistics. We analyze the storage-quality trade-off of OctaOctree by varying the number of hierarchy levels on a representative scene. Table 1 reports the maximum spatial and directional resolutions, total number of stored entries, feature storage, and the resulting reconstruction quality. Increasing the number of levels expands the complementary spatial-angular representation, providing finer spatial detail at high levels while preserving higher directional resolution at coarse levels. This generally improves image quality, but also increases storage cost. The results show that OctaOctree offers a controllable trade-off between memory usage and reconstruction fidelity.

Table 1. Statistics of OctaOctree on a representative scene with varying levels and resolutions. We report the maximum spatial/directional resolutions, number of total entries, feature storage, and image quality.

Levels	Max Res.	Total entries	Storage	MAPE	LPIPS
5	$2^6/2^5$	734,275	11.2 MB	0.178	0.106
6	$2^7/2^6$	3,526,245	53.8 MB	0.149	0.096
7	$2^8/2^7$	16,957,768	258.8 MB	0.125	0.092
8	$2^9/2^8$	79,375,000	1211 MB	0.091	0.084



Fig. 2. Visualization of ray-traced disparity and predicted parallax correction. The learned value is not a geometric hit distance, but a local reprojection field optimized for radiance reconstruction, which stabilizes neighboring octahedral lookups near reflection boundaries.

Predicted parallax correction visualization. We visualize the predicted parallax correction value for a representative scene (see Figure 2). The predicted parallax exhibits medium-frequency variation across the scene, which suggests that it adaptively distributes neighboring directional queries into different entries on the octahedral map. Meanwhile, smooth parallax distribution avoids abrupt changes in the reflected contents.